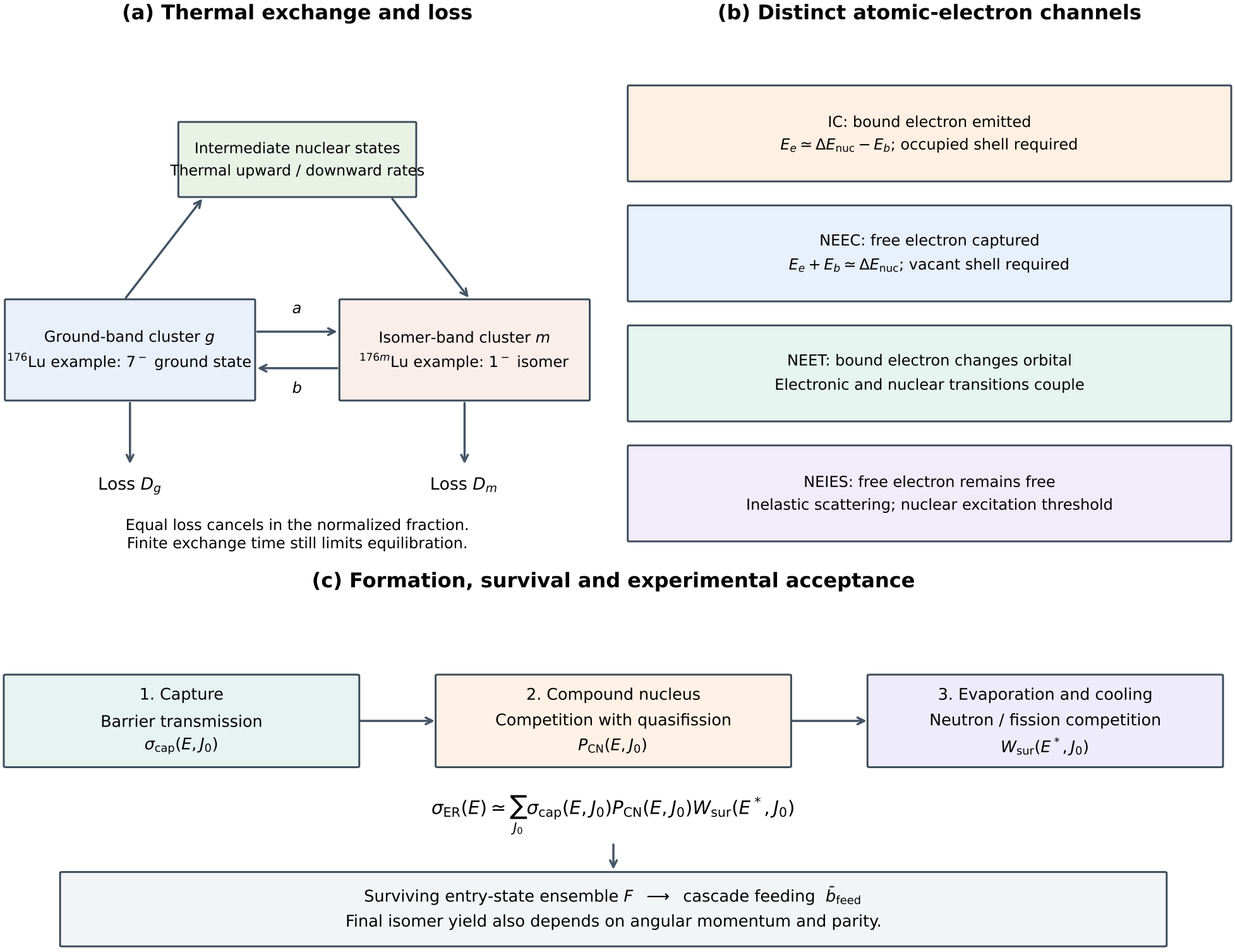
(a) Thermal exchange and loss
Intermediate nuclear states
Thermal upward / downward rates
Ground-band cluster g
^{176}Lu example: 7^- ground state
a
b
Isomer-band cluster m
^{176m}Lu example: 1^- isomer
Loss D_g
Loss D_m
Equal loss cancels in the normalized fraction.
Finite exchange time still limits equilibration.
(b) Distinct atomic-electron channels
IC: bound electron emitted
$E_e \simeq \Delta E_{\mathrm{nuc}} - E_b$; occupied shell required
NEEC: free electron captured
$E_e + E_b \simeq \Delta E_{\mathrm{nuc}}$; vacant shell required
NEET: bound electron changes orbital
Electronic and nuclear transitions couple
NEIES: free electron remains free
Inelastic scattering; nuclear excitation threshold
(c) Formation, survival and experimental acceptance
1. Capture
Barrier transmission
$\sigma_{\mathrm{cap}}(E, J_0)$
2. Compound nucleus
Competition with quasifission
$P_{\mathrm{CN}}(E, J_0)$
3. Evaporation and cooling
Neutron / fission competition
$W_{\mathrm{sur}}(E^*, J_0)$
$\sigma_{\mathrm{ER}}(E) \simeq \sum_{J_0} \sigma_{\mathrm{cap}}(E, J_0) P_{\mathrm{CN}}(E, J_0) W_{\mathrm{sur}}(E^*, J_0)$
Surviving entry-state ensemble $F \longrightarrow$ cascade feeding $\bar{b}_{\mathrm{feed}}$
Final isomer yield also depends on angular momentum and parity.
Fixed flight time and a finite observation window give a conditional decay acceptance.
Cold-state hindrance, hot-nucleus survival and detected yield are different quantities.
Schematic only: box positions and arrows do not encode energies, rates or measured branching fractions.